

2027 Proposal for North American Beacon System

This is a brief outline of the meteor observation beacon system based on SARA proposals of Sufitchi (2015) and Wilson (2016). Wilson's paper is attached.

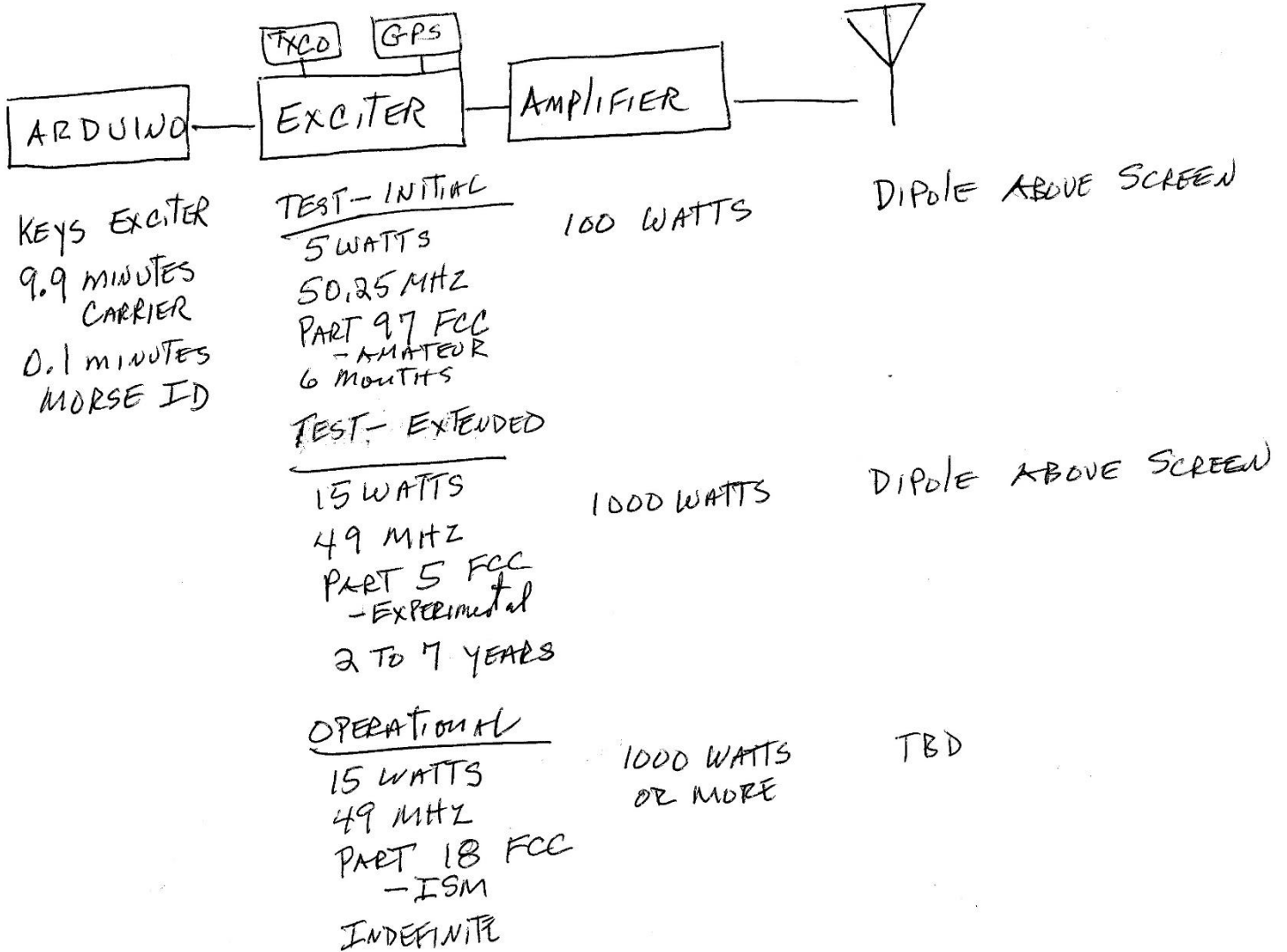
Sufitchi (N2YO), Wilson (W0AIR), and Cline (N0RQV) have obtained equipment needed to establish two 50 MHz beacon transmitters. Transmitter locations and station operators need to be identified.

Concept: Three-phase development

- Initial Test
 - Activate two 100 watt CW beacons under FCC Part 97 (Amateur Radio) regulations on the 6 meter (50 MHz) band.
 - Beacons transmit a continuous signal for 597 seconds followed by a three second Morse code identifier, then repeat.
 - FCC limits Part 97 beacons to 100 watts
 - FCC requires identification at least every 10 minutes
 - Part 97 requires electromagnetic radiation health hazards assessment, but at this low power level, safety standards are easily met.
 - Can be located at home of a volunteer for local control and emergency shut-down
- Extended Test
 - Increase power to 500 to 1000 watts
 - Continuous transmission. No signal break for identification required.
 - Operate under FCC Part 5 License (Experimental Radio Service)
 - Licenses are normally for two years
 - May be renewed for a single five year period with detailed justification
 - Nominal frequency: 48-49 MHz or as authorized by FCC
 - 48 MHz is the lowest frequency that existing equipment can be readily modified to cover
 - Limiting factor is the 1000 watt power amplifier. Currently available are two Harris Platinum amplifiers tuned for TV Channel 2.
 - With 500+ watts, RFI and safety considerations for electromagnetic health risk become a factor requiring engineering study and careful location of antennas.
 - Local or remote control
 - Ideal location would be college, university, or research organization, perhaps rooftop
- Operational System
 - 1000 watts unless higher power is needed.
 - Harris Platinum amplifiers can be used in parallel to attain desired power
 - Higher power requires very careful site planning for safety and RFI mitigation
 - Operate under FCC Part 18 (Industrial, Scientific, Medical) regulations
 - Continuous Wave (cw) unless another waveform is determined to be better
 - Local or remote control
 - Ideal location would be secure, remote property of college, university, or research organization

Reception would be by a large number of independent organizations and individuals sending reports to a central repository for analysis.

General Design



Equipment on Hand

Two 6 meter ICOM transceivers

Two Harris Platinum 1000 watt amplifiers for TV Channel 2 with power supplies

- Need retuning for 50 MHz
- Instructions available. See details at <https://www.n1rwy.org/?p=706>



Harris Platinum VHF amplifier

A Survey of Signal Sources for Future Meteor Observations

by Bascombe J. Wilson, WØAIR

SARA Eastern Conference July 2016

Overview

The study of meteors is important for many scientific purposes, details of which have been discussed in several previous SARA Conference Proceedings and Journal articles, including Sufitchi (June 2015 Proceedings) and Nelson (February and December 2015 Journals), among others. The Nelson article, in particular, details several options for setting up meteor detection systems. The present article is intended to elaborate on useful signal sources in addition to those explored by Sufitchi and Nelson in their excellent presentations.

The study of meteors in the ionosphere often uses a bistatic radar system, where meteor-scattered signals are received and analyzed at a receiver site 500-1000 kilometers distant from the transmitter. Frequencies around 40 MHz are optimal to illuminate meteors and ionized meteor trails 60-120 kilometers above the earth while avoiding direct signal reception. Frequencies above 150 MHz do not reflect well from meteor trails. A key feature for those on the receiving end is that somebody else pays for and maintains the expensive transmitters and associated licenses. To put meteor effects in context of larger ionospheric studies, Dr. Terry Bullett presents an outstanding overview of the physics of the ionosphere and upper level dynamics at http://lisn.igp.gob.pe/files/first_ionosonde_school/bullet_HF_sounding.pdf

From about 1948 until 2010, there were useful radio transmitters in virtually every direction throughout most of the world, including military radars, megawatt television transmitters and a variety of utility services. However, most high-powered military signals began disappearing in North America with the end of the Cold War, leaving analog television carrier signals on frequencies in the range of TV Channels 2-6 as the most useful and viable signal sources.

Between the years 2004-2010, major U.S. and Canadian TV stations converted from analog transmissions to digital, with greatly reduced power levels on their digital pilot carrier signals, which are usually about 310 KHz inside the channel authorization.. Stations that had previously radiated an effective power of a megawatt on their analog carrier could attain the same viewer service with just a few kilowatts on the digital pilot carrier, which is used as a frequency reference for the digital waveform, where most of the transmitted energy is now concentrated. At the same time, many stations traded their lower frequencies for UHF channel authorizations for economic reasons. On December 31, 2015, nearly all of the remaining analog TV transmitters in Mexico went silent.

While a few low power analog TV transmitters are still on the air in North America, regulators in the US, Canada and Mexico indicate that all have been ordered to transition to digital emissions and that no waivers are expected beyond the year 2020.

Most of Europe and Asia has experienced the same transition to digital television. Fortunately, France continues to operate a space surveillance bistatic radar system on 143.050 MHz that is extremely useful for

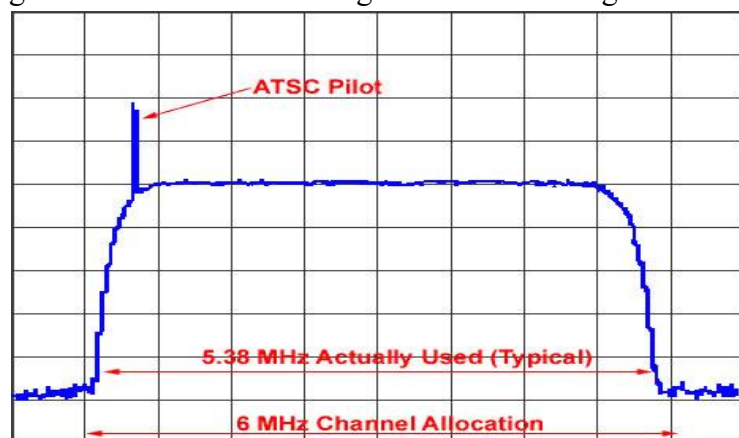
meteor observations. While a few analog TV transmitters remain on the air, virtually all are scheduled for closure in the near future. In their papers, Sufitchi and Nelson address techniques for using both analog and digital television carriers for meteor detection in North America.

This study summarizes those findings and surveys additional signals that may be useful, particularly for receiver locations where digital TV pilot carrier signals do not provide adequate illumination of meteor trails. Additionally, this study includes some important signal sources for meteor receiver stations in Europe and Africa.

Digital Television (DTV) Signals

As Nelson illustrated, DTV pilot carrier signals can be just as useful as analog carrier signals, however in most cases, useful strength of digital signals is extremely diminished because the transmitted power allocated to the DTV pilot carriers is only a small fraction of that previously used for analog carriers. Additionally, many TV transmissions have been relocated to UHF broadcast frequencies, even when the TV station retains its traditional low-channel moniker for marketing purposes. In other words, a station announcing that it is “Channel Two” is not necessarily transmitting on the Channel 2 broadcast frequency of 54 MHz. In many cases, the actual transmitter frequency is in the high UHF range, but home televisions still display the station as “Channel Two.”

Meteor-scatter reception is particularly challenging for observers in or near urban areas. For example, the author recently conducted a channel survey in Phoenix, Arizona, and could not find a single DTV signal that gave meteor returns. Strong local broadcast signals were either blocking everything else on the frequency or



scattered signals were below the ambient noise level. However, local stations and those 100-200 kilometers distant are ideal for tracking aircraft and satellites, so if one can't find the meteors, at least there is something interesting to watch.

Figure 1. Digital TV signal waveform, with pilot carrier showing as strong peak at left.
Image courtesy of Bill Horger

Meteor scatter from DTV signals is often out there; however, the challenge is finding it. There is no substitute for a fairly frequent scan of TV Channels 2-13 to determine which channels provide the best meteor scatter signals for your location. While simple dipole antennas were sufficient in the past, in the future it may be necessary to install a Yagi antenna with two or three elements to increase signal gain to the front while attenuating signals from other directions.

Currently, the Federal Communications Commission provides a searchable online database of all U.S., Canadian and Mexican television transmitters, showing actual transmit frequency, effective radiated power, and location of the transmitter. That database can be accessed at: <https://www.fcc.gov/media/television/tv-query>.

At the Little Thompson Observatory in Berthoud, Colorado, we currently obtain useful meteor scatter from both east and west DTV pilot carriers on Channel 2 (54.310 MHz). No useful analog carriers currently remain viable for us, although for years we were blessed with megawatt signals from Juarez, Mexico and Calgary, Canada.

While DTV in Europe provides an even greater challenge due to channel congestion, there are two outstanding signal sources available to many receiver stations on the continent: The French GRAVES radar system and the Belgian BRAMS meteor scatter system.

GRAVES Radar

GRAVES (Grand Réseau Adapté à la Veille Spatiale) is a French space surveillance radar transmitting a carrier signal on 143.050 MHz, with about 10,000 watts effective radiated power. The transmitter site is at Broyes-les-Pesmes (near Dijon) with site coordinates of 47°20'52"N 5°30'59"E. The primary signal is normally beamed to



the south, in an arc reaching generally from 090 degrees to 270 degrees as shown in Figure 2. Some reports indicate that the beam may be steered to the north during contingencies, but this has not been positively confirmed.

Figure 2. GRAVES transmitter pattern, showing signal backscatter to the UK

Illustration courtesy of PEØSAT and ONERA Aerospace

Laboratory

Meteor forward scatter from the GRAVES system should be very useful throughout southern Europe, Africa and much of the Middle East, with backscatter serving northern Europe and the UK. During meteor showers, it may even be possible for forward scatter to be received in the Caribbean.

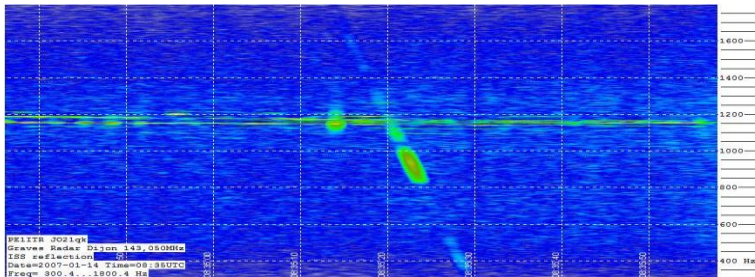


Figure 3. Example of GRAVES signal backscatter from the International Space Station as received at PE1ITR in the Netherlands.

Image courtesy of PE1ITR

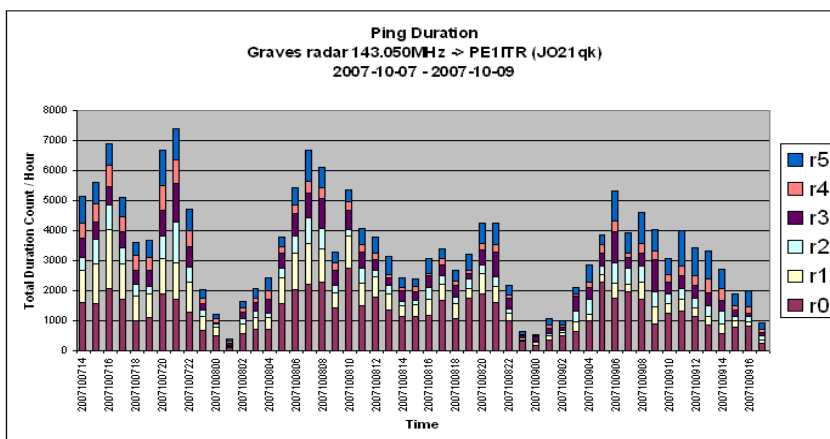


Figure 4. Total signal returns from GRAVES as recorded at PE1ITR in the Netherlands.

Image courtesy of PE1ITR

BRAMS Radio Meteor System

Belgian Radio Meteor Stations (BRAMS) transmit on 49.97 MHz, with 150 watts effective radiated power as a project of the Belgian Institute for Space Aeronomy (BISA), as part of the Solar-Terrestrial Centre of Excellence (STCE). Several transmitters are operated by Belgian radio amateurs or groups of amateur astronomers. There are two full-time beacons and numerous part-time beacons. The full-time stations are located in Leper (Western Belgium) and Dourbes (Southern Belgium). The Dourbes transmitter is located at 50,0972° N, 4,5847° E. It emits a CW circularly polarized signal at a frequency of 49.97 MHz with a constant power of 150W.

Despite the low power of this system, meteor scatter is reliably received over a 400-1000 kilometer area, primarily because of the ideal frequency selection and lack of interference on that channel. Dr. David Morgan of the British Astronomical Association has published an excellent guide on the use of BRAMS and other European signal sources, which is available at http://www.britastro.org/radio/projects/Detection_of_meteors_by_RADAR.pdf

The BRAMS system has demonstrated that a well-designed, low power beacon system can contribute much to meteor studies and can be a very cost effective alternative to megawatt radar and television broadcast

transmitters. Amateur radio astronomers and university observatories throughout Europe and the UK report useful meteor scatter reception using BRAMS signals.

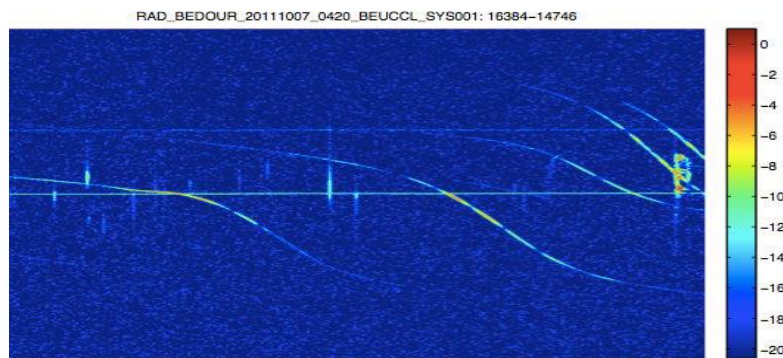


Image courtesy of Tom Roelandts

Other European Signal Sources

Amateur radio astronomer Philip Penrith, GOISW, reports on attempts at meteor forward scatter from the following European signal sources:

32.550 MHz, SkiYMET, Andenes Meteor Radar beacon in Norway with 12 KW*

32.550 MHz and 53.500 MHz, Juliusruh Meteor Radar beacon in Germany with 12 KW

36.200 MHz, Collm Meteor Radar, near Leipzig in Germany with 10 KW

39.025-39.175 MHz, Numerous commercial meteor burst communications utility stations in the UK, estimated at 100 watts

39.00-39.200 MHz, Numerous commercial meteor burst communications utility stations throughout Europe, estimated at 100 watts.

46.400 MHz, 46.950 MHz, 46.975 MHz, Meteor burst communications in the UK, 170 KW

46.500 MHz, NERC MST Weather Radar Facility in Wales, with up to 160 KW

53.500 MHz Numerous VHF radars operating from Germany, Norway and Sweden, sometimes in CW mode, other times as pulsed radars.

Figure 5. Spectrogram from BRAMS signal, showing aircraft and meteor reflections as recorded by Tom Roelandts

*None of the power levels mentioned in this paragraph were referenced to a measurement standard, so it is unclear whether the levels refer to Peak Envelope Power, Average Power, Effective Radiated Power, or some other measurement. .

A wealth of additional information about UK and European meteor detection is available at Philip's site, which is <http://www.qsl.net/g0isw/g0iswradioastronomy.htm> .

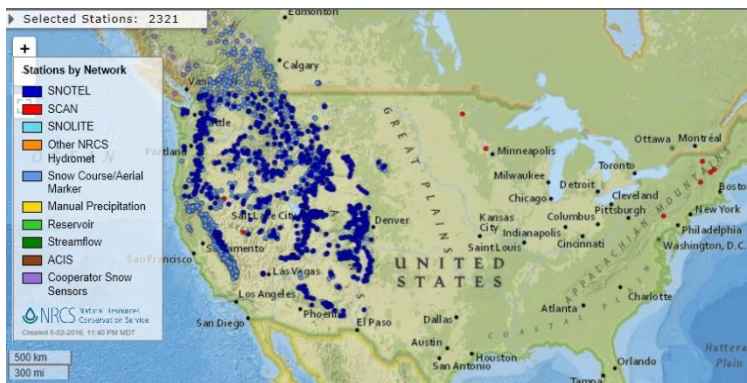
In North America, there are also some low power but useful signals that can be used to supplement DTV carriers, as well as novel applications to use FM broadcast station transmissions and amateur radio beacons on the 50 MHz and 144 MHz bands. First, the low power utility signals.

SNOTEL and SCAN in North America

SNOTEL is a system of snowpack sensors operated by the Natural Resources Conservation Service (NRCS) of the United States Department of Agriculture. There are about 730 SNOTEL (or snow telemetry) sites in 11 Western states, including Alaska. The sites are generally located in remote watersheds from which meteor burst transmissions are used to send data to central receiver sites.

SCAN is the Soil Climate Analysis Network with stations generally located in the Eastern U.S., having some overlap with SNOTEL. A few remote stations continue to use meteor burst communications, but most others now have terrestrial data links.

NRCS recently announced that the Master Stations for SCAN in the Eastern states may close at any time as remote sites transition to cellular data service. Until then, it may be possible to make use of the 40.670 MHz carrier from the Missouri master station, which will likely be the last to close. Current information is being posted to the Google Radiometeors discussion group at:



<https://groups.google.com/forum/#!forum/radiometeors>.

Figure 6. Map of NRCS SNOTEL and SCAN remote transmitters.
Graphic Courtesy of NRCS.

Master Stations for both SCAN and SNOTEL send out pilot carriers that trigger data bursts from monitoring sites whenever a meteor scatter path permits two-way communications. One strategy would be to monitor meteor scatter from the pilot carrier on 40.670 MHz. SNOTEL master stations are located near at Boise, Idaho and at Dugway Proving Grounds, Utah.

Navigational Aids for Aviation

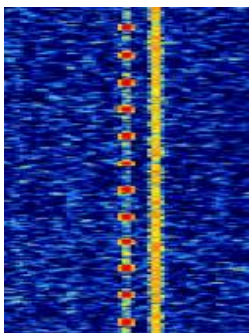
Navigational Aids (NAVAIDS) may provide some useful signal sources, particularly those designed as long range beacons. Here is a brief summary of NAVAIDS systems:

Instrument Landing System (ILS). ILS transmitters send a radio beam directly down the centerline of many airfield runways. Beginning in WW-II, ILS was developed as a precision landing aid, theoretically capable of guiding pilots to a perfect landing even with greatly reduced visibility. By 1947, ILS was an international standard and most major airfields installed transmitters on main runways. ILS is a fairly complex network of systems operating together on VHF and UHF frequencies to provide pilots with azimuth heading toward a runway and the correct descent angle needed for touchdown. Only those ILS signals with the potential for meteor scatter will be discussed.

The **ILS Localizer** sends a continuous VHF signal down the centerline of the runway, aiming toward incoming aircraft. The frequency of localizers is within the 108.10-117.95 MHz air navigation band. Localizer signals are highly directional, with several hundred watts of effective radiated power (ERP). The Localizer is designed to provide general navigational guidance 36 kilometers from the runway, and highly accurate guidance at 18 kilometers. The author has detected some meteor scatter from localizers about 300 kilometers distant. The frequency of localizer transmitters can be found on navigational charts or any one of several online flight planning sites. Just search for “**skyvector ILS LOC Pittsburgh**” or any other city to obtain ILS Localizer frequencies (abbreviated LOC on aviation charts). The best meteor scatter strategy is to find an airport ILS localizer pointing in your general direction. Remember that an ILS Localizer for Runway 09 (meaning an aircraft’s landing heading of 090 degrees) is beaming its signal toward incoming aircraft, so the transmitted signal is aimed at 270 degrees (magnetic). The Localizer signal is a continuous carrier with a modulated Morse Code identifier consisting usually of four alpha characters. Meteor scatter signals will likely be weak, but with persistence and a good airport choice, you should see some results.

Marker Beacons transmit on 75.0 MHz and beam their signals toward the zenith, usually from corner reflector antennas. Their purpose is to signal the pilot that the aircraft is passing a specific point on the ground at a known distance from the runway. There may be up to three Marker Beacons along the course leading to a runway, with each transmitting a different audio tone on the 75 MHz common channel. The most distant beacon is usually not more than 11 kilometers from the runway. Instruments in the cockpit tell the pilot which beacon is being passed, and thus, exact distance to the runway. A good visual presentation of the system is at: https://en.wikipedia.org/wiki/Marker_beacon

Marker Beacon transmissions are an AM carrier modulated signal with a beeping tone that sounds much like a telephone circuits-busy tone or an alternating dot-dash pattern, depending on location of the beacon.



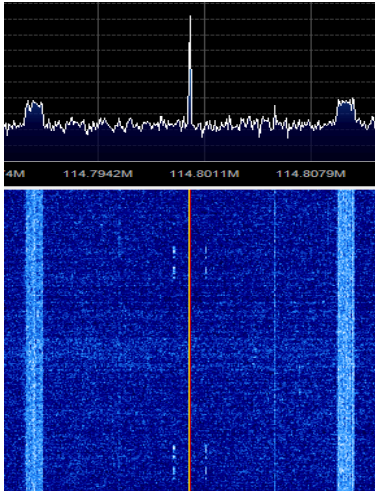
Additionally, the frequency of the audio tone is distinctly different for outer, middle and inner marker beacons as they are titled. Because there are so many 75 MHz beacons on the air, it may be relatively easy to locate signals. Signal reflections from aircraft are usually easy to detect, but the author has detected only a few meteors using Marker Beacons, likely due to relatively low ERP and the fact that most of the transmitted energy is directed to the zenith.

Figure 7. Marker Beacon signature on SpectrumLab.

VOR, or VHF Omni Directional Radio Range, may be the most promising NAVAIDS signal source for meteor scatter; however, in the U.S. there is a transition plan to discontinue the service, or at least greatly reduce the number of transmitters because of the reliability of GPS navigation. VOR was developed in 1937, and has been in use around the world since 1950. In the year 2000, there were over 3,000 VOR transmitters

around the world, with 1033 of them in the U.S. Currently, the FAA lists 967 operational VOR sites, with planned reduction to about 450 sites in 2020. Europe and the UK are conducting similar reductions..

VOR transmitters operate with 50 KHz spacing from 108.00 to 117.95 MHz, and power levels range from 250 Watts to 3 Kilowatts. The most useful VORs for meteor scatter are the ones designed for long range navigation. The FAA classification for VORs is Terminal VOR, with relatively low power to guide arriving or departing aircraft within about 25 nautical miles; Low Altitude VOR with medium power to guide aircraft below 18,000 feet altitude within 40 nautical miles; and High Altitude VOR with high power to guide aircraft up to 45,000



feet altitude, out to 113 nautical miles, but pilots report usable signals to about 200 miles. Since these altitudes and ranges are based on FAA specifications, the original English units used universally in aviation were not converted to Metric to preserve original context.

The VOR waveform is complex, with both phase modulation and amplitude modulation components. The VOR identifies itself by tone modulated Morse Code of the AM component, usually with three alpha characters transmitted slowly, visible on waterfall displays.

Figure 8. VOR signal signature on SDR#. Note the two Phase Modulation components, the carrier (red center line) and AM modulated Morse Code identification tones.

The AM carrier of High Altitude VORs may be strong enough to be scattered from prominent meteor trails. One strategy might be to pick a state in the favored reception direction at a distance of 200-500 kilometers and do an internet search for **“worldaerodata VOR Michigan”**. This should provide a table listing all the locations, frequencies and Morse IDs of VORs within that state, as well as other NAVAID information. The High Altitude VORs will be identified on charts and listings as “H.”

If you prefer map format, you may go to

http://www.faa.gov/air_traffic/flight_info/aeronav/digital_products/ifr/

Time Signal Transmissions on 20 and 25 MHz

Some observers report success receiving meteor scatter from WWV and WWVH time signal stations on 20 MHz, although this is below the ideal lower frequency for meteor observations. Nevertheless, reflections from aircraft and satellites are readily apparent. John Kraus famously tracked the reentry of Sputnik in 1958 using scattered signals of WWV, which was transmitting at that time from Maryland, and being received at the Ohio State University radio astronomy observatory in Columbus, Ohio. The author’s receiving station is within groundwave distance of WWV and has never observed meteor scatter using the signal, but aircraft and satellite returns literally fill the SpectrumLab waterfall. WWV transmits an omnidirectional signal from Fort Collins, Colorado, on 20 MHz with 2500 Watts, and is currently experimenting with occasional transmission on 25 MHz at 1000 Watts. Full technical information is available at: <http://www.nist.gov/pml/div688/grp40/wwv.cfm>

Amateur Radio VHF / UHF Beacons on the Six Meter Band (50 MHz) and to a lesser extent the Two Meter Band (144 MHz) offer some promise for meteor detection. In fact, two-way meteor bounce communications is a very popular mode of operation within the amateur radio community. There are two technical challenges: Amateur Radio beacons are generally low power and they usually transmit only on-and-off keyed Morse Code (FCC mode A1), as opposed to a continuous carrier signal, however there are some exceptions. Almost no beacon transmits above 100 Watts PEP, and a very large number are in the 5-20 watt range. On-and-off keying does not provide the steady carrier needed to detect brief meteor bursts, but reflections from more persistent meteor trails may be received.

Amateur radio operators who are most successful at meteor bounce communications usually transmit power levels near the legal limit for hams (1500 Watts PEP) into antenna arrays with considerable forward gain, often resulting in ERP levels of 10,000 to 20,000 Watts or more. Many of these stations as well as those with lower power communicate using software developed by Dr. Joe Taylor, K1JT, which maximizes very weak signal readability. While there is considerable literature and extensive field experience on using the family of K1JT digital modes for communications, the point-to-point nature of these amateur exchanges does not lend itself well to long term monitoring for meteor scatter by third parties. Nevertheless, these signal sources may prove to be valuable as easier to use and more opportunistic signals disappear. A map of North American amateur radio beacons can be found at <http://www.k9mu.com/map/> and a worldwide list is at <http://www.keele.ac.uk/depts/por/50.htm>

FM Broadcast Transmitters in the 88-108 MHz range were the subject of a novel and highly successful investigation in Dr. Brian Rachford's presentation "Using high-bandwidth FM broadcast band SDR recordings for meteors," at the 2016 SARA Western Conference. While the use of FM broadcast signals for meteor research is not new, Dr. Rachford demonstrated a technique for automatically counting and recording reflections of FM broadcast stations from meteor trails, even at a receiving site where the FM band is nearly saturated with local stations. Dr. Rachford, Astronomy Program Chair at Embry-Riddle Aeronautical University in Prescott, Arizona, provides more information about his work at <http://mercury.pr.erau.edu/~rachf7ac/>

Direct Emissions from Meteors and Meteor Trails—Not Reflected Signals. Moving to the other end of the electromagnetic spectrum, Dr. David Morgan has written a very informative paper on VLF emissions generated by meteors themselves:

http://www.britastro.org/radio/downloads/The_Generation_of_VLF_Emissions_by_Meteors.pdf

While beyond the scope of this paper, the intriguing nature of those emissions warrants further study by the amateur radio astronomy community.

A Modest Proposal: North American Beacon Network for Meteor Observation

The BRAMS system in Europe appears to provide exactly the type of signals needed for meteor studies. As legacy signal sources continue to disappear, it may be time to consider such a purpose-designed system for North America. With the likely availability of frequencies in the 40-50 MHz range for experimental licensing, and the possibility of operating a continuous carrier beacon network under amateur radio rules in the Six Meter Band, such a system should not be expensive to develop and maintain. BRAMS is successful with an ERP of 150 Watts, which could easily be provided by surplus utility transmitters or even ham radio equipment. As a starting point for discussion, the following might be considered:

A few transmitters in North America based on BRAMS Model could cover the continent

Such a network could create partnership opportunities for research institutions, amateurs and associations possibly including SARA

Cost would be modest – Utility would be very high

Key Online Sources for Additional Information:

<http://www.amsmeteors.org/cameras/>

<http://www.rmoh.org>

<http://www.cloudbait.com/projects/allskycamera.html>

<http://www.metrec.org/>

<http://www.merriott-astro.co.uk>

http://sonotaco.com/e_index.html

<http://cams.seti.org/>

<http://brams.aeronomy.be/>

www.onera.fr/synindex-en/graves-radar.html

http://www.britastro.org/radio/projects/BRAMS_Meteor_Radar.pdf

<http://www.britastro.org/radio/>

<http://www.eracnet.org>

http://isr.sri.com/sondre_brochure.pdf

<http://www.astron.nl/radio-observatory/radio-observatory>

Lists of Television Transmitters in North America

https://en.wikipedia.org/wiki/Lists_of_television_stations_in_North_America

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